

数值分析第二十次作业

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1. 证明对于任意参数 α , 下列 Runge-Kutta 格式是二阶的:

$$\begin{cases} y_{n+1} = y_n + \frac{h}{2}(K_2 + K_3), \\ K_1 = f(t_n, y_n), \\ K_2 = f(t_n + \alpha h, y_n + \alpha h K_1), \\ K_3 = f(t_n + (1 - \alpha)h, y_n + (1 - \alpha)h K_2). \end{cases}$$

解答: 记 $f(t_n, y_n) = f_n$, 由 Taylor 展开可得截断误差

$$\begin{aligned} & y(t_{n+1}) - y(t_n) - \frac{h}{2}(K_2 + K_3) \\ &= hf_n + \frac{h^2}{2}(\partial_t f_n + \partial_y f_n \cdot f_n) \\ &\quad - \frac{h}{2}(f_n + \alpha h \partial_t f_n + \alpha h \partial_y f_n \cdot f_n + f_n + (1 - \alpha)h \partial_t f_n + (1 - \alpha)h \partial_y f_n \cdot f_n) + \mathcal{O}(h^3) \\ &= \mathcal{O}(h^3), \end{aligned}$$

故该 Runge-Kutta 格式是二阶的. □

2. 对具有 Butcher 表 $(\mathbf{A}, \mathbf{b}, \mathbf{c})$ 的 s 级 Runge-Kutta 方法, 证明其放大因子为

$$R(z) = 1 + z\mathbf{b}^T(\mathbf{I} - z\mathbf{A})^{-1} \cdot \mathbf{1} = \frac{\det(\mathbf{I} - z\mathbf{A} + z\mathbf{1} \cdot \mathbf{b}^T)}{\det(\mathbf{I} - z\mathbf{A})}.$$

解答: 由降阶公式可得

$$\begin{aligned} \frac{\det(\mathbf{I} - z\mathbf{A} + z\mathbf{1} \cdot \mathbf{b}^T)}{\det(\mathbf{I} - z\mathbf{A})} &= \det((\mathbf{I} - z\mathbf{A})^{-1}(\mathbf{I} - z\mathbf{A} + z\mathbf{1} \cdot \mathbf{b}^T)) \\ &= \det(\mathbf{I} + z(\mathbf{I} - z\mathbf{A})^{-1}\mathbf{1} \cdot \mathbf{b}^T) \\ &= 1 + z\mathbf{b}^T(\mathbf{I} - z\mathbf{A})^{-1} \cdot \mathbf{1}. \end{aligned}$$
□